

Activation Sensitivity Analysis for DLMS

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Issue with Llada SINQ Quantization

- The ff_out inside the blocks was not being quantized.
- The goal is to quantize the ff_out inside the block but keep the last ff_out unquantized

```
LLaDAModelLM(
  (model): LLaDAModel(
    (transformer): ModuleDict(
      (wte): Embedding(126464, 4096)
      (emb_drop): Dropout(p=0.0, inplace=False)
      (ln_f): RMSLayerNorm()
      (blocks): ModuleList(
        (0-31): 32 x LLaDALlamaBlock(
          (dropout): Dropout(p=0.0, inplace=False)
          (act): SiLU()
          (attn_out): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cuda, W_q.device=cpu)
          (ff_out): Linear(in_features=12288, out_features=4096, bias=False)
          (rotary_emb): RotaryEmbedding()
          (attn_norm): RMSLayerNorm()
          (ff_norm): RMSLayerNorm()
          (q_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cuda, W_q.device=cpu)
          (k_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cuda, W_q.device=cpu)
          (v_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cuda, W_q.device=cpu)
          (ff_proj): SINQLinear(in_features=4096, out_features=12288, bias=False, device=cuda, W_q.device=cpu)
          (up_proj): SINQLinear(in_features=4096, out_features=12288, bias=False, device=cuda, W_q.device=cpu)
        )
      )
      (ff_out): Linear(in_features=4096, out_features=126464, bias=False)
    )
  )
)
```

Evaluation Of Absmax Dynamic Quantization

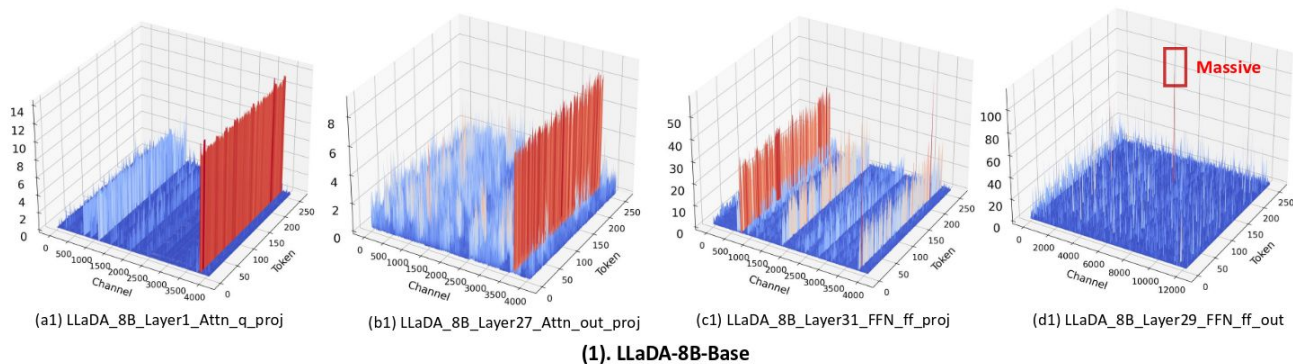
- The matmul operations are still conducted in full precision, the activations are quantized and then dequantized before matrix multiplication

4-bit quantization of Dream-7B-Base and LLaDA-8B-Base. The numbers indicate the % of correct and incorrect answers of the baseline model that changed.

Model	ARC-Challenge	ARC-Easy	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base (SINQ W4A16)	3.16	2.90	1.52	5.68	8.22	0.92
LLaDA-8B-Base (SINQ W4A8)	4.69	4.08	1.84	8.61	9.98	1.21
LLaDA-8B-Base (SINQ W4A4)	34.98	44.28	27.47	46.57	51.89	27.74
Dream-7B-Base (SINQ W4A16)	6.23	3.41	3.37	5.68	—	—
Dream-7B-Base (SINQ W4A8)	6.31	3.96	3.54	7.97	—	—
Dream-7B-Base (SINQ W4A4)	46.33	60.86	33.08	37.97	—	—

Outliers in LLaDA Activations

- Massive Outliers tend to occur in the second linear layer of the feed-forward network (ff_out)



H. Lin *et al.*, “Quantization Meets dLLMs: A Systematic Study of Post-training Quantization for Diffusion LLMs,” Oct. 15, 2025, *arXiv*: arXiv:2508.14896.

Effect of Clipping Activations On Flip Rates

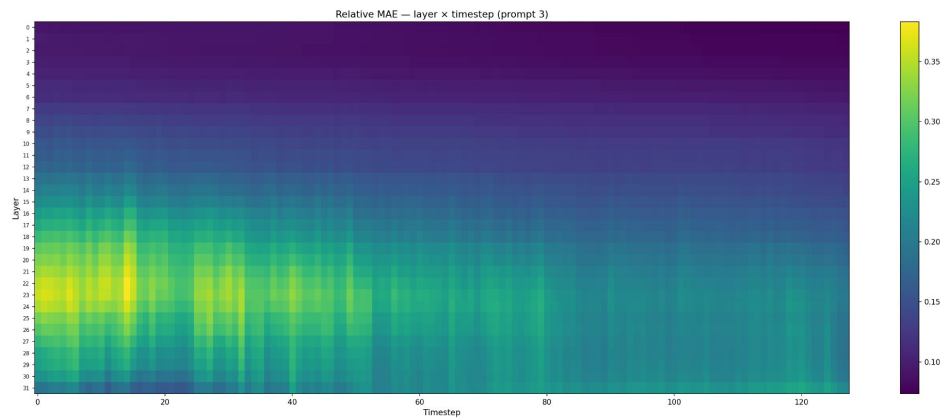
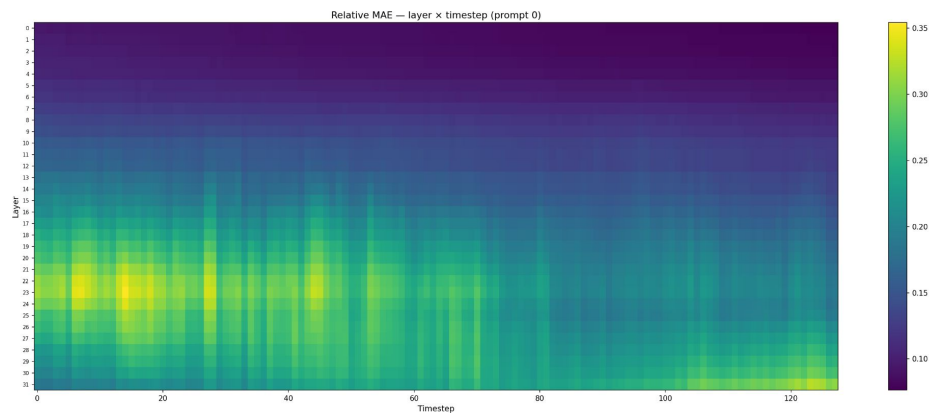
- Clipping activations destroys the results for W4A8 case
- The 99.99% clipping of activations slightly improve the W4A4 results which proves that massive outliers are affecting the quality of W4A4 model

Model	ARC-Challenge	ARC-Easy	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base (SINQ W4A16)	3.16	2.90	1.52	5.68	8.22	0.92
LLaDA-8B-Base (SINQ W4A8)	4.69	4.08	1.84	8.61	9.98	1.21
LLaDA-8B-Base (SINQ W4A8 clip 0.99)	40.10	56.22	36.12	46.25	52.74	—
LLaDA-8B-Base (SINQ W4A8 clip 0.999)	37.88	48.52	29.38	43.72	52.87	—
LLaDA-8B-Base (SINQ W4A8 clip 0.9999)	7.85	6.69	3.70	13.58	20.93	—
LLaDA-8B-Base (SINQ W4A4)	34.98	44.28	27.47	46.57	51.89	27.74
LLaDA-8B-Base (SINQ W4A4 clip 0.99)	39.16	54.88	35.69	46.01	51.36	—
LLaDA-8B-Base (SINQ W4A4 clip 0.999)	37.63	49.08	30.19	46.88	52.80	—
LLaDA-8B-Base (SINQ W4A4 clip 0.9999)	32.85	37.24	23.67	45.70	51.69	—
Dream-7B-Base (SINQ W4A16)	6.23	3.41	3.37	5.68	—	—
Dream-7B-Base (SINQ W4A8)	6.31	3.96	3.54	7.97	—	—
Dream-7B-Base (SINQ W4A8 clip 0.99)	45.91	60.90	32.59	34.25	—	—
Dream-7B-Base (SINQ W4A8 clip 0.999)	45.39	57.24	31.28	33.71	—	—
Dream-7B-Base (SINQ W4A8 clip 0.9999)	27.04	25.97	15.51	24.55	—	—
Dream-7B-Base (SINQ W4A4)	46.33	60.86	33.08	37.97	—	—
Dream-7B-Base (SINQ W4A4 clip 0.99)	46.93	61.40	33.51	33.86	—	—
Dream-7B-Base (SINQ W4A4 clip 0.999)	45.56	60.31	32.21	35.68	—	—
Dream-7B-Base (SINQ W4A4 clip 0.9999)	46.84	60.01	32.59	36.62	—	—

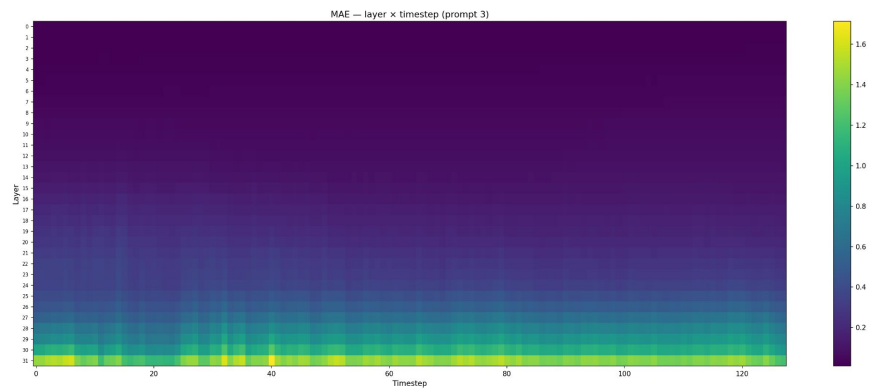
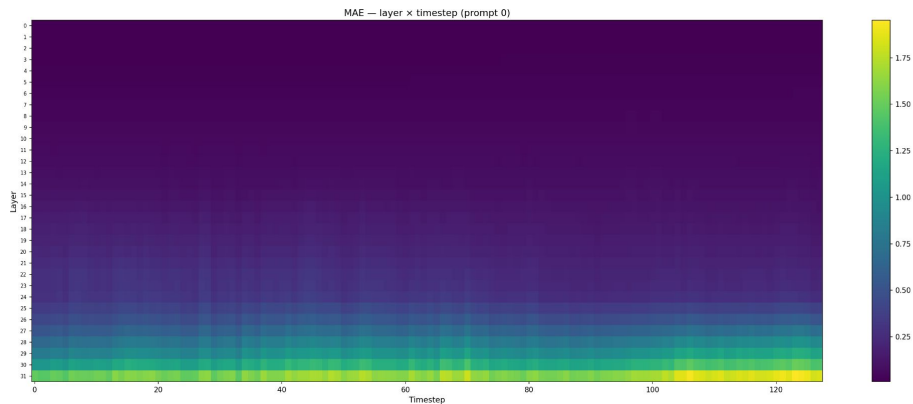
Analyzing Activation Sensitivity

- The fp16 Llada model is compared against the SINQ quantized model on several prompts across all layers and timesteps
- The goal is to understand the sensitivity of activations to this quantization to help us in designing a proper activation quantization technique for DLMs

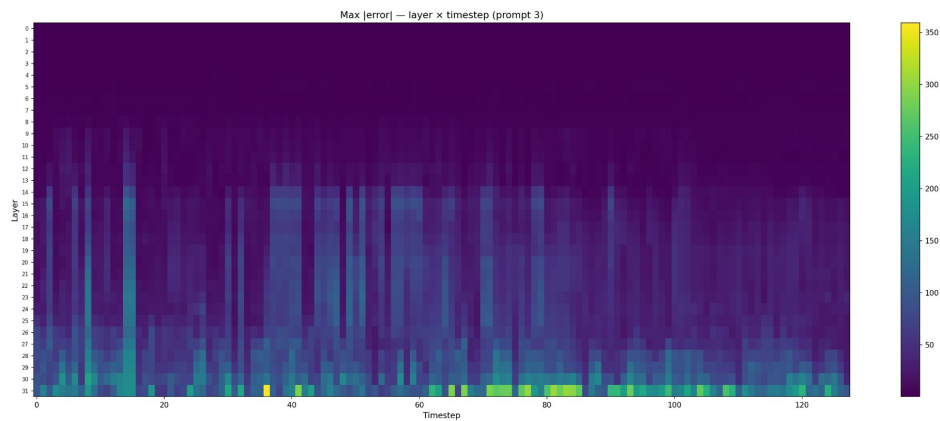
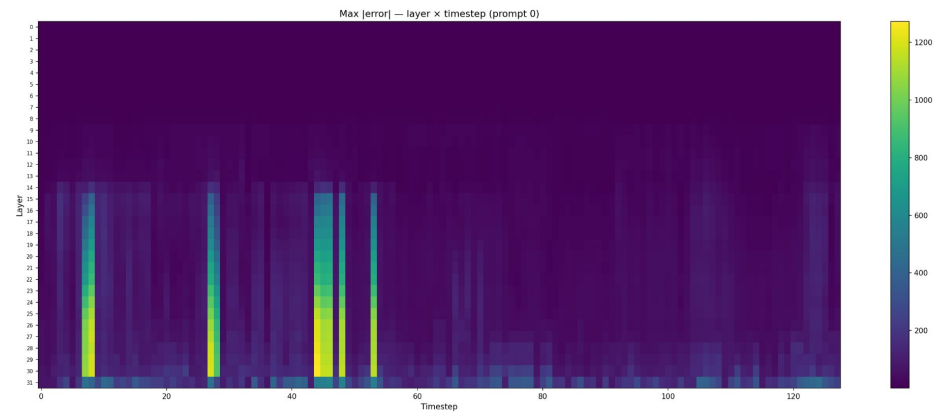
Relative Mean Absolute Error in Llada Model



Mean Absolute Error



Max Absolute Error



Plan for next week

- Apply different activation quantization techniques across different layers and timesteps based on the current analysis and compare results to baseline

